Intro to Machine Learning — Milestone 2 Report—

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1 Introduction

Milestone 2 (MS2) returns to the Gaming and Mental Health dataset (2000 synthetic samples, 13 behavioural features) introduced in Milestone 1 (MS1) and reopens the same two questions: predict the continuous addiction score (regression) and the 3-class addiction level {Low, Medium, High} (classification). MS1 concluded that the signal was near-linear in the 13 z-scored features. MS2 puts that conclusion under stress by adding a non-linear parametric model — a Multi-Layer Perceptron (MLP) trained from scratch with back-propagation — and a fundamentally different family — K-Means used as a voting classifier. Class imbalance (Low 68.9%, Medium 27.9%, High 3.2%) keeps macro-F1 as the primary classification metric; mean squared error (MSE) is the regression metric. Hyperparameters are selected by 5-fold cross-validation (CV).

2 Methodology

2.1 Data & Preprocessing

We z-score normalize the 13 features using *training-set* statistics only. Crucially, we shuffle the training data with a fixed random seed *before* taking an 80/20 train/validation split, removing the leakage risk that an ordered split would otherwise introduce. The validation split is used for hyperparameter selection only; the final test set (N=400) is held out until the very last evaluation.

2.2 Validation protocol

For classification we use stratified 5-fold CV — the High class has only $n\approx 41$ training samples, so non-stratified folds risk empty validation slices. We report mean and standard deviation across the five folds. Hyperparameters are selected by the highest CV macro-F1 (and not by classification error rate); for K-Means we also report the within-cluster sum of squares (inertia) to compare init strategies on their native objective.

3 Methods & Results

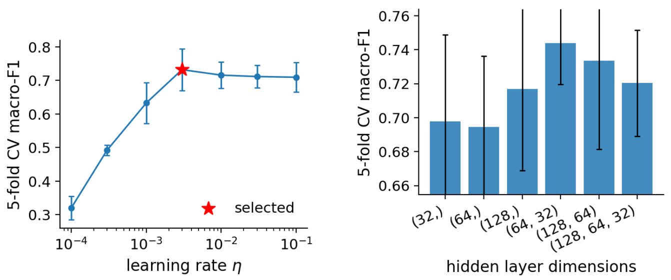
3.1 Multi-Layer Perceptron

We implement a fully-connected MLP with arbitrary depth, ReLU or sigmoid activations, and an output layer chosen by task: Softmax with Cross-Entropy (CE) loss for classification, Identity with MSE for regression. Forward and backward passes are coded by hand; the optimiser is mini-batch Stochastic Gradient Descent (SGD) with optional heavy-ball momentum (β). Weights use Glorot/Xavier initialisation ($\sigma=\sqrt{2/(\text{fan-in}+\text{fan-out})}$) to avoid gradient vanishing.

Hyperparameter sweeps. We swept the learning rate η on a log grid from 10^{-4} to 10^{-1} (Fig. 1a) and six hidden architectures from (32) to (128, 64, 32) (Fig. 1b). The lr curve is sharply peaked: F1 collapses for $\eta \lesssim 10^{-3}$ (too few effective steps in 40 epochs) and degrades mildly for $\eta \gtrsim 10^{-2}$ (noisier updates), with the best CV F1 at η=$3{\cdot}10^{-3}$. The architecture sweep shows a real but moderate effect: single-layer nets cluster around CV F1 0.70 and the two-layer $(64, 32)$ wins at **0.744**, with the three-layer $(128, 64, 32)$ dropping back to 0.72 (harder to optimise in 40 epochs). The signal is therefore mostly linear, but one extra hidden layer buys a small genuine gain. We select $(64, 32)$.

Selected configuration. Hidden layers $(64, 32)$ ReLU, η=$3{\cdot}10^{-3}$, momentum β=0.9, batch size 32, 40 epochs (CE+Softmax for classification, MSE+Identity for regression).

Ablations (5-fold CV F1 around the selected config). Three knobs are cliffs: momentum (F1 0.48/0.56/0.74 for β=0/0.5/0.9), activation (Sigmoid 0.51 vs ReLU 0.69), and training budget (epochs 10: 0.54; batch 128: 0.55 — both under-train in 40 epochs). Two knobs are within fold noise: batch size 8 vs 32, and CE+Softmax vs MSE-on-onehot as a sanity check. We keep CE+Softmax for the final classifier because it is the matched probabilistic objective.



(a) Learning rate η (log-x); red star = selected.

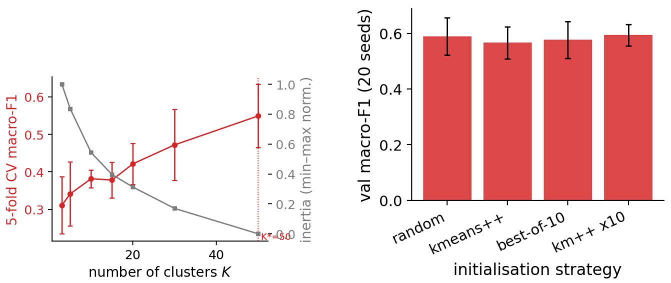
(b) Hidden-layer architecture; $(64, 32)$ peaks.

Figure 1: MLP sweeps (5-fold stratified CV macro-F1, error bars are fold std).

3.2 K-Means voting classifier

K-Means partitions the training set into K clusters by iteratively assigning points to their nearest centre (Euclidean distance) and re-computing centres as cluster means. To use it as a classifier we attach a label to each cluster by majority vote over its training points, then classify a new point as the label of its nearest cluster. We implement two extensions: K-Means++ seeding (each new centre sampled with probability proportional to its squared distance to the nearest existing centre) and best-of-N restarts (keep the lowest-inertia run out of N).

Choosing K. The number of clusters K controls a strong tension on this dataset (Fig. 2a). Inertia decreases mono-



(a) K sweep: CV macro-F1 (red) and normalised inertia (grey). F1 keeps rising over the swept range; selected K=50.

(b) Init strategy at K=50, 20 seeds each: better inertia does *not* translate to better F1.

Figure 2: K-Means sweeps on the train split.

tonically with K (no clear elbow), and *F1* keeps rising across the whole sweep, peaking at K=50 (CV F1 0.55). The reason is class imbalance: with K=3 a majority vote inside each cluster collapses to the global majority (Low), giving accuracy $\approx$71% but macro-F1 0.31. Increasing K

creates smaller, locally purer clusters where Medium and High samples can dominate, lifting minority-class recall. We select $K{=}50$ (largest K on the grid, with an average training-fold cluster size of about 25 points, avoiding an obvious memorisation regime).

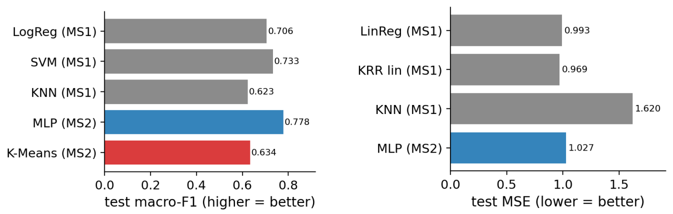
Restarts cut variance; K-Means++ alone does not help. Sweeping four init strategies at $K{=}50$ (20 seeds each, Fig. 2b): K-Means++ alone neither lowers inertia ($\bar{I}{=}8571$ vs random 8543) nor F1 (0.57 vs 0.59); ten restarts do the real work (inertia σ: 27→16 with km++×10). All four mean F1s land within ~ 0.03 of each other — inside the seed-to-seed std — so the inertia and voting objectives are misaligned on imbalanced data.

4 Discussion & Conclusion

4.1 MS2 vs MS1: which family wins?

Fig. 3 and Table 2 consolidate every selected MS1 and MS2 model on the same held-out test set. Three take-aways stand out.

MLP beats every MS1 classifier. On classification, the MLP reaches test macro-F1 **0.778**, a clear +0.045 over MS1's best (Linear SVM at 0.733). This refines, rather than contradicts, the MS1 conclusion that the signal is near-linear: one extra hidden layer plus class-aware Softmax+CE buys a real but bounded gain.



(a) Classification: test macro-F1 (higher is better).

(b) Regression: test MSE (lower is better).

Figure 3: Final test-set comparison. Grey = MS1 models, blue = MS2 MLP, red = MS2 K-Means.

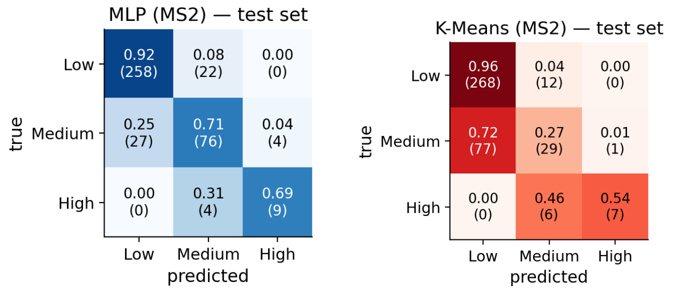
Table 1: Per-class precision / recall / F1 on the official test set for the two MS2 classifiers, complementing Fig. 4. The MLP–K-Means F1 gap is concentrated on the Medium class.

Method	Class	Prec.	Recall	F1
MLP	Low	0.91	0.92	0.91
	Medium	0.75	0.71	0.73
	High	0.69	0.69	0.69
K-Means	Low	0.78	0.96	0.86
	Medium	0.62	0.27	0.38
	High	0.88	0.54	0.67

The gain over SVM lands on the minority classes (per-class F1 in Table 1, especially High) — exactly the bottleneck MS1 flagged. The architecture sweep attributes +0.04 CV F1 to the extra hidden layer; the ablation attributes the rest to momentum and training time.

MLP regression matches ridge within noise. A separate η sweep on the regression task bottoms out at $\eta{=}3{\cdot}10^{-4}$ (CV MSE 1.13 vs 1.33 at the classification-best $3{\cdot}10^{-3}$); with this task-specific η the MLP gives test MSE **1.027** vs ridge (0.993) — a 0.034 gap, within the ~ 0.11 fold std. K-Means could be turned into a regressor (mean cluster target), but matching ridge on a continuous target would need $K{\gg}50$ and is outside scope.

K-Means is comparable to k-NN but well below parametric methods. The voting classifier reaches test F1 0.634, marginally above MS1's k-NN (0.623) but inside the $\sim$0.07 seed std (Fig. 2b), and 0.14 below the MLP. The confusion matrices (Fig. 4) localise the gap: K-Means handles Low (recall 0.96) and even captures High at high precision (0.88, recall 0.54) but *collapses on Medium* (recall 0.27, F1 0.38): 63/107 Medium samples are voted as Low. The MLP is balanced across all three classes (recall = 0.92/0.71/0.69). The MLP–K-Means gap is essentially a Medium-class story: unsupervised partitioning of an imbalanced, mostly-linear feature space does not align with the class boundaries CE optimises directly.



(a) MLP: balanced.

(b) K-Means: collapses Medium.

Figure 4: Test-set confusion matrices (row-normalised recall; counts in parentheses).

Table 2: Selected configurations and test-set performance across the two milestones. Reg: MSE (lower = better); Clf: accuracy / macro-F1 (higher = better). Best of each task in bold. *km++×10* denotes K-Means++ seeding with 10 restarts.

Method	Task	Config	Test
LinReg (MS1)	reg	CF ridge $\lambda{=}10$	MSE 0.993
KRR (MS1)	reg	linear kernel, $\lambda{=}1$	MSE **0.969**
KNN (MS1)	reg	$k{=}17$, cosine, weighted	MSE 1.62
MLP (MS2)	reg	$(64, 32)$, $\eta{=}3{\cdot}10^{-4}$, $\beta{=}0.9$	MSE 1.027
LogReg (MS1)	clf	balanced wts, $\eta{=}0.3$, $\beta{=}0.9$	80.4% / F1 0.706
SVM (MS1)	clf	$\lambda{=}10^{-2}$, batch 32	82.3% / F1 0.733
KNN (MS1)	clf	$k{=}17$, cosine, weighted	80.1% / F1 0.623
MLP (MS2)	clf	$(64, 32)$ ReLU, Softmax+CE, $\beta{=}0.9$	85.8% / F1 **0.778**
K-Means (MS2)	clf	$K{=}50$, km++×10	76.0% / F1 0.634

4.2 Conclusion

A from-scratch MLP with $(64, 32)$ ReLU, Softmax+CE and momentum is the best classifier across both milestones (test F1 **0.778**, +0.045 over MS1's SVM); the ablation isolates momentum, ReLU and training budget as the three cliffs. MLP regression matches MS1's ridge within fold noise once its own η is used. *Seed robustness.* Over 10 final MLP refits, classification reaches F1 0.793 ± 0.021 and regression MSE 1.062 ± 0.017. K-Means trails every parametric model: K matters far more than initialisation, and the inertia objective is misaligned with the imbalanced voting objective.